Assumptions Document

Data Quality Assumptions

1. Price Field

Assumption: The `price` column in listings and calendar files contains USD amounts with `$` prefix and comma separators

Implication: Must strip `$` and `,` before converting to numeric

Validation: After cleaning, ensure no negative prices and reasonable ranges

Treatment: Negative or zero prices will be flagged as invalid and handled with explicit nulls

2. Availability Dates

Assumption: Calendar data covers 365 days from the snapshot date

Implication: Availability_365 should match sum of available calendar days

Validation: Check consistency between `availability_365` and calendar counts

Treatment: If mismatch >5%, investigate and document

3. Missing Values'

Assumption: Missing values represent unknown information rather than meaningful categories

Implication: Explicit nulls will be used rather than imputation in most cases

Validation: High null rates in review scores may indicate newer listings

Treatment: Nulls preserved as `NULL` in star schema; analysts can decide imputation strategy

4. Host Information

Assumption: Host data (response_rate, host_since) is as reported and current

Implication: Older records may have less accurate host information

Validation: Compare host_tenure calculations with host_since dates

Treatment: Host_since before snapshot date is validated; impossible dates flagged

5. Neighbourhood Mapping

Assumption: Neighbourhood names are consistent across files

Implication: Joins on neighbourhood name should be exact matches

Validation: Check for misspellings or variations (e.g., "Harlem" vs "Central Harlem")

Treatment: Fuzzy matching for small discrepancies; manual review for major differences

6. Review Scores

Assumption: Review scores are out of 10 (overall rating system)

Implication: Scores can be compared meaningfully across listings

Validation: Check score distribution for expected patterns (typically 4.0-5.0)

Treatment: Scores outside expected range flagged for investigation

7. City Snapshot Dates

Assumption: All files for a city come from the same snapshot date

Implication: Data is consistent within each city dataset

Validation: Check file timestamps and data recency

Treatment: If dates differ, note in decision log

8. Regulatory Context

Assumption: Regulatory impacts (NYC Local Law 18, Barcelona 2028 ban) are reflected in market dynamics

Implication: Analysis should examine whether these regulations correlate with observable patterns

Validation: Compare trends before/after key regulatory dates where data allows

Treatment: Regulatory analysis is qualitative and contextual, not causal

Data Processing Assumptions

9. Cross-City Harmonization

Assumption: Column names and types are similar enough across cities to harmonize

Implication: Can build unified pipeline with mapping for minor differences

Validation: Schema comparison conducted in notebook

Treatment: Differences documented in decision log

10. Star Schema Implementation

Assumption: Analytical queries are best served by dimensional model

Implication: Denormalization decisions balance query performance vs storage

Validation: Query performance tested with representative use cases

Treatment: Trade-offs documented and indexed appropriately

Business Context Assumptions

11. Calendar as Occupancy Proxy

Assumption: Unavailable nights in calendar represent booked nights

Implication: Revenue estimates = price × nights_unavailable

Validation: Compare with review counts as demand proxy

Treatment: Explicitly caveat in analysis

12. Market Definition

Assumption: Each city is a distinct market with own dynamics

Implication: Cross-city comparisons need normalization

Validation: Market concentration and supply characteristics compared

Treatment: Report suggests areas for deeper investigation

Limitations

1. **No Transaction Data:** Calendar availability is a proxy, not actual bookings
2. **Snapshot Nature:** Quarterly snapshots miss continuous dynamics
3. **Potential Scraping Artifacts:** Data quality may vary by snapshot
4. **Missing Historical Context:** Only current snapshot available
5. **Regulatory Attribution:** Cannot establish causation between regulations and market changes
6. **Platform Changes:** Airbnb policy changes not captured
7. **Guest Demographics:** No data on who books or for what purpose
8. **Competitor Context:** No data on alternative accommodations (hotels, VRBO)

Decision Log Reference

All major decisions in the project are documented in `reports/decision_log.md` with:

- Options considered - Chosen approach - Trade-offs accepted - Rationale for selection

Database Selection

- **Decision:** Switched from DuckDB to SQLite
- **Rationale:**
- SQLite has better compatibility across systems
- No need for pre-existing database files
- Built into Python standard library
- Simpler for the assessment environment
- **Trade-offs:**
- SQLite has less analytical functionality than DuckDB
- Some advanced window functions not available
- But still sufficient for all required queries